Mukesh Patel School of Technology Management & Engineering
School of Technology Management and Engineering

Course Policy Document

Course Name - (Code): Generative AI (GenAI)

<table>
<tr><td>Program and Semester: MBA Tech (Artificial Intelligence) , Semester VIII</td><td>Pre-requisite Course: Deep Learning, Advanced Techniques in Machine Learning</td></tr>
<tr><td>Academic Year: 2025-26</td><td>Credit Details:
<table>
<tr><td>L</td><td>T</td><td>P</td><td>C</td><td>H</td></tr>
<tr><td>2</td><td>0</td><td>2</td><td>3</td><td>4</td></tr>
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<tr><td>Office Hours:
Tuesday 2.00 pm to 4.00 pm
Friday: 3.00 pm to 5.00 pm</td><td>Faculty associated with the course: -
1. Dr. Vaishali Kulkarni
Vaishali.kulkarni@nmims.edu</td></tr>
<tr><td colspan="2">Pre-Course Activity:
Using GenAI, explore the following:
1. Applications of Generative AI
2. Latest Developments in LLMs specifically in multimodal LLMs
3. Identify how this course will help you in your career goals.
4. Identify the progress of Generative AI after the paper on transformer architecture was proposed in 2017.</td></tr>
<tr><td colspan="2">Course link:
General | Generative AI_MBA Tech_2025_26 | Microsoft Teams</td></tr>
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1. Introduction to the Course

1.1 Importance of the Course

1.1.1 Domain Relevance:

The Generative AI course is highly significant for BTech students specializing in Artificial Intelligence, as it bridges foundational AI knowledge with cutting-edge developments in generative technologies. It equips students with essential theoretical and practical skills to design, train, and deploy advanced generative models such as Large Language Models (LLMs)**,** Diffusion Models, and Large Vision Models (LVMs)**.** By exploring techniques like Retrieval-Augmented Generation (RAG) and Parameter-Efficient Fine-Tuning (PEFT), students gain the ability to optimize models for real-world applications across text, vision, and multimodal tasks. Moreover, the course fosters understanding of ethical, agentic, and performance evaluation aspects,

preparing students to innovate responsibly in rapidly evolving AI domains such as creative content generation, conversational AI, autonomous systems, and intelligent agents.

1.1.2 Industry Relevance:

The Generative AI course holds strong industry relevance as it aligns directly with the technologies driving the next wave of innovation across sectors such as healthcare, finance, education, entertainment, and robotics. With companies increasingly leveraging Large Language Models (LLMs)**,** Diffusion Models, and AI agents for automation, creativity, and decision-making, this course prepares students with hands-on expertise in tools and frameworks like LangChain**,** LLaMA Index, and RAG-based systems. The skills gained—such as fine-tuning large models, generating realistic data, building intelligent assistants, and evaluating AI systems—are in high demand for roles like AI engineer, data scientist, and machine learning researcher. By emphasizing both the technical depth and ethical deployment of generative systems, the course ensures graduates are industry-ready to contribute to cutting-edge AI product development and innovation.

1.2 Objectives of the Course:

This course aims to provide a comprehensive understanding of fundamental concepts, advanced architectures, and practical applications of Generative Artificial Intelligence. The course emphasizes both theoretical foundations and hands-on learning, equipping students with skills to build, evaluate, and deploy generative AI models responsibly.

2. Course Outcomes (CO), Mapping with Program Outcomes (PO), Program Specific Outcomes (PSO) and Student Outcomes (SO)

2.1 Course Outcomes

2.1.1 CO1: Understand the foundations of Generative AI including its applications.

2.1.2 CO2: Analyze various Large Language Model (LLM) architectures.

2.1.3 CO3: Apply retrieval-augmented generation (RAG), and parameter-efficient fine-tuning (PEFT) techniques to optimize AI model performance.

2.1.4 CO4: Develop Generative AI applications with Diffusion and Large Vision models.

2.1.5 CO5: Evaluate Generative AI models using quantitative metrics.

2.1.6 **CO6:** Recognize the practical considerations of Agentic AI.

2.2 Program Outcomes (PO) the course contributed to:

2.2.1 PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

2.2.2 PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

2.2.3 PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

2.2.4 PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

2.2.5 PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

2.2.6 PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

2.2.7 PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

2.2.8 PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

2.2.9 PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

2.2.10 PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

2.2.11 PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for independent and life-long learning ii) adaptability to new and

emerging technologies and iii) critical thinking in the broadest context of technological change.

2.3 Program Specific Outcomes (PSO)

2.3.1 Design solutions to various industrial problems to benefit society and the environment using Artificial Intelligence technologies through professional development activities.

2.3.2 Pursue research, innovation and entrepreneurship in emerging areas of Artificial Intelligence by integrating domain knowledge and multidisciplinary approaches.

2.4 Program Specific Outcomes (PSO): CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	2	2	3	2	1	1	1	1	1	2	2
CO2	2	3	3	2	3			1	1	1	2	3	2
CO3	1	3	2	3	3	2	1	1	1	1	1	3	2
CO4	1	2	2	2	3	2	1	1	1	1	1	3	2
CO5	1	2	3	2	3	2	1	1	1	1	1	3	2
CO6		2	3	2	3	2	1	1			1	3	1

Mapping Levels: 1- Low, 2-Medium, 3- High

3. Teaching-learning methodology

3.1 Instruction Plan

Lecture No.	**Topic**	**Teaching Method ***	**Blooms Level**	**References**	**COs mapped**	**Assessment and Evaluation**
Unit 1: Foundations of Generative AI						
1	Introduction to Generative Models,	PPT, videos	Understand	T1	CO1	Class Test-1 and TEE
2	Applications of Large Models for Vision and Language.		Understand	T1	CO1	Class Test-1 and TEE
Unit 2: Large Language Models (LLM)						
3	Introduction to LLM: difference between traditional language models and LLM	PPT	Understand	T1, R1	CO1	Lab Submission and

4	Auto regressive models		Apply	T1, R3	CO2	Class Test-1 and TEE
5	Overview of attention mechanisms, multi-head attention, and self-attention in transformer-based models.		Apply	T1, R3, Base paper on transformer, “Attention is all you need”	CO2	
6	Positional Embeddings(PE): Importance of long sequences, drawbacks of sinusoidal PEs, Rotary based PE (RoPE).		Understand, Apply	T1, R3, Base paper on transformer, “Attention is all you need”	CO2	
7	Encoder-only model: BERT (bidirectional Encoder Representation Transformer) architecture		Understand, Apply	T1, R3, Base paper on BERT	CO2	
8	Decoder-only model - Architecture of GPT (Generative Pretrained Transformer)		Understand, Apply	T1, R3, Base paper on GPT1	CO2	
9	GPT2, GPT3		Understand, Apply	T1, R3, Base papers on GPT2 and GPT3	CO2	Lab Submission and Class Test-1 and TEE
10	Encoder-Decoder model - BART		Understand, Apply	T1, R3	CO1	
11	LLM Frameworks: Langchain, LLaMA Index	PPT	Apply	T1, T2, R2, Langchain and LLaMA documentation	CO1	
12	Test 1					
Unit 3: RAG and Fine Tuning						

13	Vector Databases, High-dimensional Data Storage	PPT, Whiteboard	Analyse, Apply	T1, T2, open-source Vector database documentation	CO3	Lab Submission and Class Test-2 and TEE
14	Vector Embeddings, High-Dimensional Semantic Similarity		Analyse, Apply	T1, T2, open-source Vector database documentation	CO3	Lab Submission and Class Test-2 and TEE
15	Vector Embeddings, High-Dimensional Semantic Similarity (HNSW and LSH algorithms)		Analyse, Apply	T1, T2, open-source Vector database documentation	CO3	Lab Submission and Class Test-2 and TEE
16	Retrieval-Augmented Generation (RAG)		Apply	T1, T2	CO3	
17	Hyperparameter Tuning and Optimization, Full fine-Tuning v/s Parameter Efficient Fine Tuning (PEFT) Techniques, LoRA, QLoRA	PPT, Assignment	Understand and implement	T1, T2, research papers on Parameter Efficient Fine Tuning (PEFT) Techniques	CO3	Lab Submission and assignment and TEE
Unit 4: Generative Vision with Diffusion and Large Vision Models (LVM)						
18	Overview of LVMs (ViT), architectures, applications	PPT	Understand and Analyse	T3, research paper on ViT	CO4	Practicum and Class Test-2 and TEE

19	Overview of LVMs (CLIP), architectures, applications classification and object detection. Training and fine tuning of LVM.		Understand and Analyse	T3, research paper on CLIP	CO4	Practicum and and Class Test-2 and TEE
20	Introduction to Diffusion Models-Understanding core concept of diffusion model (DDPM)	PPT	Understand and Analyse	T3 research paper on Diffusion	CO4	Quiz and TEE
21	Test 2					
22	generate data from noise	PPT	Understand and Analyse	T3 research paper on Diffusion	CO4	Quiz and TEE
23	comparison with GAN	PPT	Understand and Analyse	T3 research paper on Diffusion	CO4	Quiz and TEE
24	Training DDPM for image generation, super resolution and denoising.	PPT	Understand and Analyse	T3 research paper on Diffusion	CO4	Quiz and TEE
Unit 5: Generative AI Evaluation						
25	Evaluation of LLM: Perplexity (Language Modelling), BLEU, METEOR	PPT	Understand	T1	CO5	Quiz and TEE
26	ROUGE (Machine Translation, Summarization), MRR (Question Answering), Recall@K (Retrieval Tasks)	PPT	Understand	T1	CO5	Quiz and TEE
27	Benchmarks: GLUE, MMLU, BigBench, HELM, OpenLLM	PPT	Understand	T1	CO5	Quiz and TEE
Unit 6: Introduction to Agentic AI and its Applications						

28	Definition and Concept of Agentic AI	PPT	Understand and implement	R3	CO6	Lab submission and TEE
29	Difference between LLMs and AI Agents Components of an AI Agent	PPT	Understand and implement	R3	CO6	
30	Ethical and Practical Considerations in Agentic AI	PPT	Understand and implement	R3	CO6	Lab Submissions and TEE

Text books and Reference books

Text Books

1. T1: "Foundations of Large Language Models", Tong Xiao and Jingbo Zhu**,** Natural Language Processing lab, North Eastern University & Niutrans Research, Jan 2025
2. T2: "Transformers for Natural Language Processing", Dennis Rothman, Packt Publishing, Jan 2021.
3. T3: "Hands-On Generative AI with Transformers and Diffusion Models", Omar Sanseviero, Pedro Cuenca, Apolinário Passos, Jonathan Whitaker, O'Reilly Media, Inc., November 2024

Reference Books

1. R1: "Build a Large Language Model (From Scratch)", Sebastian Raschka, Manning Publications, 2025
2. R2: "Generative AI with LangChain", Ben Auffarth, Packt Publishing, 2023
3. R3. " AI Agents in Action", Micheal Lanham , Manning Publication , 2025

4. Assessment and Evaluation Scheme

	Internal Continuous Assessment (ICA) (50 Marks)				
	Class Tests (20 Marks)		**Term-work (30 Marks)**		
Assessment Component	Class Test-1	Class Test-2	Lab Performance and practicum	Mini Project and Research Paper	Assignment and quiz
Marks	10	10	10	10	10

4.1 Internal Continuous Assessment (ICA) – 50 marks

Two class tests will be conducted as per the academic calendar.

Lab performance (5 marks) Continuous assessment for laboratory experiments will be conducted. **50% marks will be deducted for late submissions. (It is important**

that all lab submissions to be done before the deadline given). Discussion of your work with your peers is allowed. However, each student is expected to submit his/her original work. Submissions, which are very similar, will be marked zero.

Practicum (5 marks)

i. Group of 2-3 students will design a lab problem statement based on course contents and provide solutions for the same.
ii. This will be documented and submitted as per the timeline mentioned.
iii. Rubric is as follows
    - Unique problem definition-1Marks
    - Problem solution (algorithm/code)-2Marks
    - Write up- 2Marks

Quizzes or Assignments: The faculty will conduct the quizzes and/or assignments

Mini Project (10 marks) – Group of 2-3 students will work in a team. The faculty member should approve the topic and title of the mini project by the end of 3rd week. The mini project can be combined with any other course, wherever applicable with the approval of both the concerned faculty members. Report should be plagiarism free. Plagiarism checker tools will check the report.

Evaluation of project will be based on following rubrics

- Domain knowledge and literature review in the selected topic (2 marks)
- Implementation, performance evaluation, learning beyond classroom (5 marks)
- Report (3 marks)

4.2 Term End Examination (TEE) – (100 marks scaled down to 50)

TEE conducted at the end of the semester will be for 100 marks (3-hour duration). Marks obtained scaled down to 50. There will be 7 questions (each question will be 20 marks). Q1 will be compulsory and any 4 from the remaining to be solved.

4.3 Course Passing Criteria

4.3.1 ICA (50 marks) – No minimum marks

4.3.2 TEE (100 marks scaled to 50) – 40% required for passing

4.3.3 (ICA + TEE) (100 marks) - 50% required for passing

4.4 Assessments and Mapping to Course Outcomes

	Internal Continuous Assessment (ICA)					**Term End Examination (TEE)**
Course Outcomes	CT-1	CT-2	Lab Performance + Lab Exam /Viva	Quizzes and /or Assignments	Mini Project	TEE
CO-1	Y		Y	Y	Y	Y
CO-2	Y	Y	Y	Y	Y	Y

CO-3		Y	Y	Y	Y	Y
CO-4		Y	Y	Y	Y	Y
CO-5			Y	Y	Y	Y
CO-6			Y	Y	Y	Y

5. Laboratory details

The following 10 programming exercises will form the submission for laboratory coursework. Each programming exercise will contain 3 to 5 programs.

Expt. No.	Week No.#	List of Lab Exercises	Mapped CO
1	1	**Explore Generative Models: Concepts, Applications** Task 1: Application Mapping Task 2: Foundation Model Timeline Task 3: Explore pre-trained foundation models via no-code or low-code interfaces Task 4: Reading and comprehending the literature	CO1
2	2	**To implement a mini transformer-based model and explore foundation models.** Task 1: Comprehending research papers Task 2: Understand and implement key components of a Transformer decoder (GPT-style) including self-attention, multi-head attention, and positional embeddings	CO1
3	3	**Explore and implement BERT models.** Task 1: Comprehending the BERT paper by Google Task 2: BERT implementation using TensorFlow hub Task 3: BERT implementation using Hugging Face Task 4: Compare performances for your use case in task 2 and 3	CO2
4	4	**Comparative Analysis of LLM Architectures – BERT, GPT, BART**. Load pretrained BERT, GPT-2, and BART using Hugging Face Fine-tune: • BERT for sentiment classification. • GPT-2 for text generation. • BART for text summarization or question-answering. Evaluate and compare the behavior, outputs, and use cases of each architecture.	CO2
	5	First phase of evaluation of mini project	All COs
5	6	**Exploring basic concepts of LangChain** Task 1: Creating API Key, setting up an environment and using LangChain's model primitive Task 2: Using LangChain PromptTemplates Task 3: Structured JSON (JSON output parsing) Task 4: Knowing about LangChain	CO3
6 & 7	7	**Building LLM Applications with LangChain** • Students will form groups of 4**.**	CO3

		• Each group will choose one application from the list (or propose their own with instructor approval).
		• The application should be implemented using LangChain and, optionally, an open-source or hosted LLM (Mistral, LLaMA 2, GPT-4, etc.).
		• Use at least three LangChain components such as:
		  • PromptTemplate
		  • Chain or LCEL composition
		  • Memory (for conversational apps)
		  • Tool/Agent (for API/database integration)
		  • OutputParser (for structured responses like JSON)
		  • VectorStoreRetriever (for RAG over documents)	

6 & 7	8	Continuation of earlier week lab	CO2
8	9	**To implement and compare vector databases (ChromaDB and Weaviate) for semantic search applications** Task 1: Setup & ChromaDB Implementation Task 2: Setup & Weaviate Implementation Task 3: Comparative Analysis Task 4: Real world use case implementation (Choose One, you can work on data scrapped for any of the use cases you choose) • Semantic FAQ Search. • News Recommendation Engine. • Personal Notes / Research Paper Search. • Code Snippet Finder. Task 5: Reflection Questions	CO3
9	10	**To implement a Retrieval-Augmented Generation (RAG) pipeline using a vector database, with emphasis on chunking techniques and HNSW algorithm for nearest neighbour search.** Task 1: Dataset Preparation Task 2: Embedding & Vector Storage Task 3: Retrieval with HNSW Task 4: RAG Pipeline Task 5: Analysis & Reflection	CO3
10	11	To explore Agentic AI.	CO6
	12	Second phase of evaluation of mini project	All COs
	13	Practicum	CO4, CO5
	14	Final evaluation of Mini project	All Cos
	15	Final evaluation of Mini project	All Cos

6. Course Material

References and Lab Manuals would be uploaded on Teams by faculty every week.

7. GenAI Usage

7.1 Pre-class Activities:

- You can use AI tools to generate summaries, explanations, and examples of Generative AI concepts covered in the upcoming class.
- Utilize AI to find additional resources such as articles, papers, and tutorials relevant to the class content.

7.2 In-Class Activity:

- Engage with AI-driven interactive simulations and visualizations to understand the learning algorithms and architectures.

7.3 Lab /Assignments

- You can use AI to gather information, find research papers, and explore various methods related to the assignment topic.
- Use AI for brainstorming and generating initial ideas for assignments.

Students should not use Generative AI to:

- For creating entire code or solutions using AI without understanding and personal contribution
- Directly copying AI-generated content and passing it off as their own work**/Project**
- GenAI is permitted to be used only for the deployment of models or learning new concepts beyond the classroom. Students need to cite the references used.

7.4 Projects

- Use AI to assist in planning, design, and outlining project ideas.
- Utilize AI tools to conduct research, analyze data, and find relevant information for their projects.
- GenAI is permitted to be used only for the deployment of model or learning new concepts beyond classroom. Students need to cite the references used.

Students must **not** use Generative AI to:

1. Develop entire project**;** the core implementation and critical thinking aspects of the project must be the students' own work.
2. Use AI in ways that bypass the learning process and objectives of the course.

Activities for which Generative AI cannot Be Used

1. AI tools are not allowed during exams and quizzes to ensure academic integrity.
2. Any assignments specifically designed to assess the students' individual understanding and problem-solving skills should not involve AI-generated content.
3. Tasks that require original thought, unique problem-solving approaches, or creative content must be completed without AI assistance.

8. Academic Integrity Statement

Original work expected from students for all of the **assigned assessment work**. Copying in any form is not acceptable and will invite strict disciplinary action. Evaluation of corresponding components will be affected proportionately in such cases. Plagiarism detection software will be used to check plagiarism wherever applicable. Academic integrity is expected from students in all components of course assessment.

*** - Only Teaching Method in the Instruction Plan for the course may vary for different faculty teaching the course, rest of the Course Policy Document will not change.**